ECE 329 Introduction to Electromagnetic Fields Spring 17

University of Illinois **Waldrop, Lee, Goddard, Ilie**

Exam 3

Thursday, April 20, 2017 — 7:00-8:15 PM

Please clearly PRINT your name in CAPITAL LETTERS and circle your section in the boxes below.

Name:	Solutions
Section:	11 AM 12 Noon 1 PM 2 PM

If you want your exam returned to a **different** section, write it here:

Special Return to Section:	11 AM 12 Noon 1 PM 2 PM

This is a closed book exam and calculators/electronic devices are not allowed. Please show all your work and make sure to include your reasoning for each answer. All answers should include **units** wherever appropriate. The exam is double sided. You may use the back of the exam as scratch paper.

Problem 1 (6 points)	
Problem 2 (18 points)	
Problem 3 (27 points)	
Problem 4 (18 points)	
Problem 5 (20 points)	
Problem 6 (11 points)	
TOTAL (100 points)	

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Because calculators are not allowed, please use the following expressions for physical constants:

$$c = 3 \times 10^8 \left[\frac{m}{s}\right]$$

$$\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \left[\frac{F}{m}\right]$$

$$\mu_0 = 4\pi \times 10^{-7} \left[\frac{H}{m}\right]$$

$$\eta_0 = 120\pi \ [\Omega]$$

and leave your answers in the form of fractions rather than decimals.

A Table for Your Reference

23 Imperfect dielectrics, good conductors

	Condition	β	α	$\lvert\eta\rvert$	τ	$\lambda = \frac{2\pi}{\beta}$	$\delta = \frac{1}{\alpha}$
Perfect dielectric	$\sigma = 0$	$\omega\sqrt{\epsilon\mu}$	0	$\sqrt{\frac{\mu}{\epsilon}}$	0	$\frac{2\pi}{\omega\sqrt{\epsilon\mu}}$	∞
Imperfect dielectric	$\frac{\sigma}{\omega\epsilon} \ll 1$	$\sim \omega\sqrt{\epsilon\mu}$	$\beta\frac{1}{2}\frac{\sigma}{\omega\epsilon} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\epsilon}}$	$\sim \sqrt{\frac{\mu}{\epsilon}}$	$\sim \frac{\sigma}{2\omega\epsilon}$	$\sim \frac{2\pi}{\omega\sqrt{\epsilon\mu}}$	$\frac{2}{\sigma}\sqrt{\frac{\epsilon}{\mu}}$
Good conductor	$\frac{\sigma}{\omega\epsilon} \gg 1$	$\sim \sqrt{\pi f\mu\sigma}$	$\sim \sqrt{\pi f\mu\sigma}$	$\sqrt{\frac{\omega\mu}{\sigma}}$	$45°$	$\sim \frac{2\pi}{\sqrt{\pi f\mu\sigma}}$	$\sim \frac{1}{\sqrt{\pi f\mu\sigma}}$
Perfect conductor	$\sigma = \infty$	∞	∞	0	-	0	0

- The table above summarizes TEM wave parameters in homogeneous conducting media where the propagation velocity

$$v_p = \frac{\omega}{\beta}$$

x-polarized phasor

$$\tilde{\mathbf{E}} = \hat{x}E_o e^{\mp\alpha z} e^{\mp j\beta z}$$

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1. (6 points) TRUE or FALSE questions (2 points each). Circle either **TRUE** or **FALSE**. Note that some of these questions require some calculation to get the right answer, but you do not need to show your work to get credit. No partial credit is given on these questions.

   a) TRUE or FALSE: Copper behaves like a good conductor at all frequencies. Fails at high frequency

   b) TRUE or FALSE: The pair of fields $\mathbf{E} = \eta_0 H_0 \sin(\beta x)\sin(\omega t)\hat{y}$ and $\mathbf{H} = H_0 \cos(\beta x)\cos(\omega t)\hat{z}$ fields satisfy Maxwell's equations in free space. This is a standing wave

   c) TRUE or FALSE: Standing waves propagate zero net energy. $S \neq 0$ but $\langle S \rangle = 0$

2. (18 points) Multiple choice questions (3 points each). Circle only **ONE** letter next to the **ONE** correct statement. Most questions require some calculation to get the right answer, but you do not need to show your work to get credit. No partial credit.

   (i) Which statement is true about TEM waves propagating in a good conductor?

   a) The wave propagates a distance of several wavelengths before the field amplitude decays to its $\frac{1}{e}$ value.

   b) The wave propagation velocity is the same for all frequencies.

   c) The **E** and **H** fields are in phase.

   d) The time averaged power flow is constant along the propagation direction.

   e) None of these.

   (ii) Consider a co-sinusoidal plane TEM wave propagating as a guided wave between two infinitely long, parallel conducting plates with **E** polarized along $\hat{x}$ and **H** along $\hat{y}$. The plates are at $x = 0$ and $x = d$ and they extend to a distance W along the y-axis with $W \gg d$. Which statements is true?

   a) At a given value of z and a fixed moment in time, the voltage difference between the two plates depends on the path taken between the two plates.

   b) At a given value of z and a time when $E_x > 0$, negative charge is induced on the $x = 0$ plate while positive charge is induced on the $x = d$ plate.

   c) At a given value of z and a time when $E_x > 0$, positive charge is induced on both the $x = 0$ and the $x = d$ plates.

   d) Current flow will be induced in the plates along the $\pm\hat{y}$ directions.

   e) None of these.

   (iii) The magnetic field of a linearly polarized wave in a lossless magnetic material having the permitivity of free space ϵ_0 and intrinsic impedence of $\eta = 1500\Omega$ is $\mathbf{H} = 5Re[e^{j(\omega t - \beta y)}]\hat{x}\,[A/m]$. The corresponding electric field in the time domain in units of $[V/m]$ is:

   a) $\mathbf{E} = -7500cos(\omega t + \beta y)\hat{z}$

   b) $\mathbf{E} = 7500cos(\omega t - \beta y)\hat{z}$

   c) $\mathbf{E} = \frac{1}{1500}cos(\omega t + \beta y)\hat{z}$

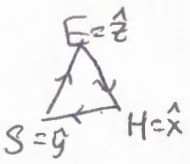
   d) $\mathbf{E} = -\frac{1}{1500}cos(\omega t - \beta y)\hat{z}$

   e) None of these.

$H = 5\cos(\omega t - \beta y)\hat{x}$

$E = \eta H$

$= 7500\cos(\omega t - \beta y)\hat{z}$



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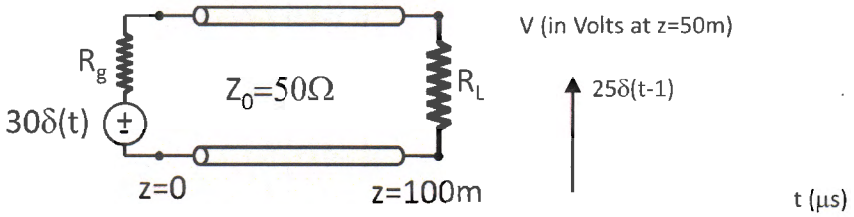
(iv) The reflection coefficient Γ for the electric field of a TEM wave propagating from a material with properties $\epsilon = 4\epsilon_0$, $\mu = \mu_0$, and $\sigma = 400\omega\epsilon_0$ into free space is approximately:

a) 0
b) $\frac{1}{3}$
c) $-\frac{1}{3}$
d) 1
e) -1
f) ∞
g) Cannot be determined.

$\sigma >> \omega\varepsilon$ so initial wave is in a good conductor whereby η_1 is approaching zero

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \rightarrow +1$$

(v) A pulse generator with unknown internal resistance sends a delta function pulse at $t = 0$s down a transmission line circuit that is terminated by an unknown resistive load. The voltage is measured at $z = 50$m (i.e. the midpoint of the line) and a single pulse is observed after $1\,\mu s$. No other pulses are observed.



The internal resistance of the generator in ohms is:

a) 0
b) 10
c) 30
d) 50
e) 100
f) ∞
g) None of these.

$$\tau = \frac{50}{50+R_g} \cdot 30 = 25$$

(vi) For the circuit and voltage measurement of the previous problem, the load resistance is:

a) 0
b) 10
c) 30
d) 50
e) 100
f) ∞
g) None of these.

TL is impedance matched because no reflection

$\Rightarrow R_L = Z_0$

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3. (27 points) In conducting media, plane TEM wave parameters γ and η satisfy

$$\gamma\eta = j\omega\mu \text{ and } \frac{\gamma}{\eta} = \sigma + j\omega\epsilon$$

The electric field of such a plane wave propagating in a non-magnetic material ($\mu = \mu_0$) is given by:

$$\mathbf{E} = 3e^{-3x}\cos(8\pi 10^6 t - 4x + 60°)\hat{z} \text{ V/m}$$

Noting physical constants given on page 2 and that $\tan^{-1}(\frac{4}{3}) = 53°$, determine the following quantities:

a) (3 pts) The wave's attenuation constant α, wave number β, and propagation constant γ

Your Answer (include appropriate units):

$\alpha =$ 3 Np/m $\beta =$ 4 rad/m $\gamma =$ 3+4j rad/m

b) (4 pts) The amplitude and phase of the propagation constant γ

$|\gamma| = \sqrt{\alpha^2+\beta^2} = \sqrt{3^2+4^2} = 5$ $\quad \angle\gamma = \tan^{-1}(\frac{4}{3}) = 53°$

Your Answer (include appropriate units):

$|\gamma| =$ 5 $\quad angle(\gamma) =$ 53°

c) (6 pts) The wavelength λ, linear frequency f, and phase velocity v_p of the wave

$\lambda = \frac{2\pi}{\beta} = \frac{\pi}{2}$ $\quad \omega = 8\pi\cdot 10^6 = 2\pi f \Rightarrow f = 4\cdot 10^6$ $\quad v = \frac{\omega}{\beta} = \frac{8\pi\cdot 10^6}{4} = 2\pi\cdot 10^6$

Your Answer (include appropriate units):

$\lambda =$ $\pi/2$ m $\quad f =$ 4 MHz $\quad v_p =$ $2\pi\cdot 10^6$ m/s

d) (6 pts) The amplitude and phase of the intrisic impedance η of the medium

$\eta = \frac{j\omega\mu}{\gamma} \Rightarrow |\eta| = \left|\frac{\omega\mu}{\gamma}\right| = \frac{8\pi\cdot 10^6 \cdot 4\pi\cdot 10^{-7}}{5} = \frac{32\pi^2}{50}$

Because $\eta\gamma = j\omega\mu$, $\angle\eta + \angle\gamma = \angle j = 90°$

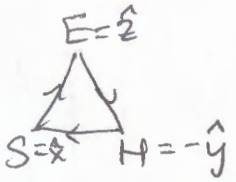
$\angle\eta + \angle\gamma = 90°$

$\Rightarrow \tau = 90 - 53 = 37°$

Your Answer (include appropriate units):

$|\eta| =$ $\frac{32\pi^2}{50}$ or $\frac{64\pi^2}{100}$ or $0.64\pi^2$ Ω $\quad \tau = angle(\eta) =$ 37°

e) (8 pts) The associated magnetic field $\mathbf{H}$.



Your Answer (include vector direction and appropriate units):

$\mathbf{H} =$ $\frac{3}{|\eta|} e^{-3x} \cos(8\pi\cdot 10^6 t - 4x + 60° - 37°)(-\hat{y})$ A/m

23°

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4. (18 points) Consider a plane wave with $\mathbf{E} = E_x \cos(\omega t - \beta z)\hat{x} + E_y \sin(\omega t - \beta z + \phi)\hat{y}$ traveling through a lossless medium.

a) (8 pts) What are the phasors $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$?

$\sin x = \cos(x - \frac{\pi}{2})$

$E = E_x \cos(\omega t - \beta z)\hat{x} + E_y \cos(\omega t - \beta z - \frac{\pi}{2} + \phi)\hat{y}$

$H = \frac{E_x}{\eta}\cos(\omega t - \beta z)\hat{y} + \frac{E_y}{\eta}\cos(\omega t - \beta z - \frac{\pi}{2} + \phi)(-\hat{x})$

Your Answer (include vector directions and appropriate units):

$\tilde{\mathbf{E}} = E_x e^{-j\beta z}\hat{x} + E_y e^{-j\beta z} e^{j(\phi - \frac{\pi}{2})}\hat{y}$ V/m

$\tilde{\mathbf{H}} = \frac{E_x}{\eta} e^{-j\beta z}\hat{y} + \frac{E_y}{\eta} e^{-j\beta z} e^{j(\phi - \pi/2)}(-\hat{x})$ A/m

b) (6 pts) What is the time averaged Poynting vector?

$\langle S \rangle = \frac{1}{2}\mathrm{Re}[\tilde{E} \times \tilde{H}^*]$

Because $\hat{x} \times \hat{x} = 0$ and $\hat{y} \times \hat{y} = 0$ the cross terms in the cross product are zero. Thus, only the first and last terms survive

$\tilde{E} \times \tilde{H}^* = (E_x e^{-j\beta z}\hat{x}) \times (\frac{E_x}{\eta} e^{+j\beta z}\hat{y}) + (E_y e^{-j\beta z} e^{j(\phi - \frac{\pi}{2})}\hat{y}) \times (\frac{E_y}{\eta} e^{+j\beta z} e^{-j(\phi - \frac{\pi}{2})}(-\hat{x}))$

first terms; last terms

$= \frac{E_x^2}{\eta}\hat{z} + \frac{E_y^2}{\eta}\hat{z}$

Your Answer (include vector directions and appropriate units):

$\langle \mathbf{S} \rangle = \frac{1}{2}\eta(E_x^2 + E_y^2)\hat{z}$ W/m^2

c) (4 pts) Circle the condition under which the wave left handed circularly polarized (LHCP):

A. $E_x = E_y$ and $\phi = \frac{\pi}{2}$

B. $E_x = -E_y$ and $\phi = \frac{\pi}{2}$

C. $E_x = E_y$ and $\phi = 0$

D. $E_x = -E_y$ and $\phi = 0$

E. None of these.

For RHCP, Ahead E x Behind E = prop. direction and $\hat{x} \times \hat{y} = \hat{z}$ = prop direction

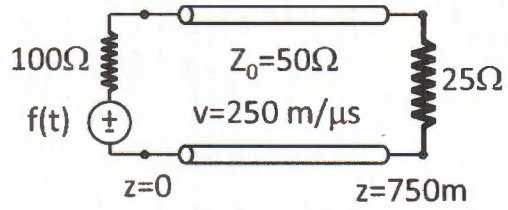
So LHCP $\Rightarrow$ $\cos(\omega t - \beta z - \frac{\pi}{2} + \phi)$ needs to be ahead to get ahead E x Behind E = $-$prop. dir.

$\Rightarrow \phi - \frac{\pi}{2} = \frac{\pi}{2} \Rightarrow \phi = \pi$

or $\phi = 0$ and $E_x = -E_y$

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5. (20 points) Consider a transmission line with characteristic impedance $Z_o = 50\,\Omega$, length $l = 750$ m, $v = 250\ \frac{m}{\mu s}$. A voltage source with internal resistance $R_g = 100\,\Omega$ is connected at the $z = 0$ end, while a load resistance $R_L = 25\,\Omega$ terminates the line at $z = l$.



a) (6 pts) Compute the transmission line parameters τ_g, Γ_g, and Γ_L for voltage:

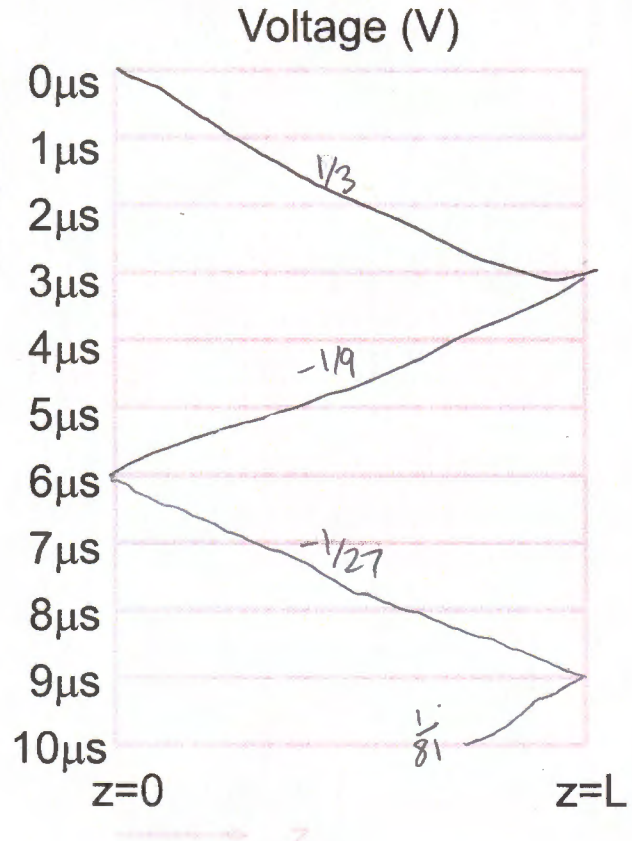
$\tau_g = \frac{50}{100+50} = \frac{1}{3}$ $\quad \Gamma_g = \frac{100-50}{100+50} = \frac{1}{3}$ $\quad \Gamma_L = \frac{25-50}{25+50} = -\frac{1}{3}$

Your Answer:

$\tau_g =$ 1/3 $\quad \Gamma_g =$ 1/3 $\quad \Gamma_L =$ -1/3

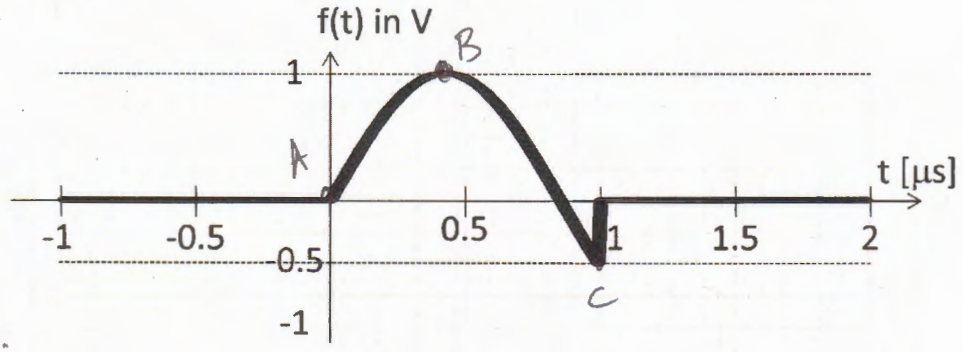
b) (4 pts) Construct the voltage bounce diagram for $f(t) = u(t)$V, where $u(t)$ is the unit-step function. Label the voltage amplitude on each bounce (you do not need to write out the functional forms for the voltage waves).

$T = \frac{l}{v} = \frac{750\,m}{250\,m/\mu s} = 3\mu s$



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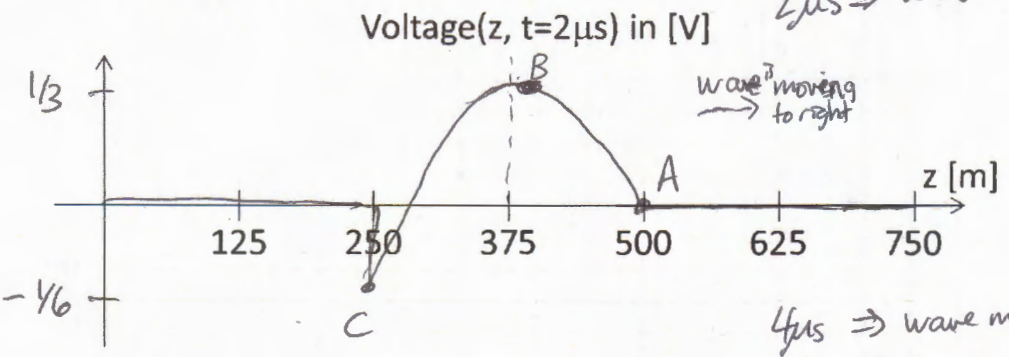
c) (10 pts) Consider the voltage source given by $f(t) = sin(\omega t) rect\left(\frac{t-0.5\,\mu s}{T}\right)$ where $T = 1\,\mu s$ and $\omega/2\pi = \frac{7}{12}$ MHz as shown below. The dotted lines show that the maximum and minimum for $f(t)$ are $+1$ and -0.5, respectively.



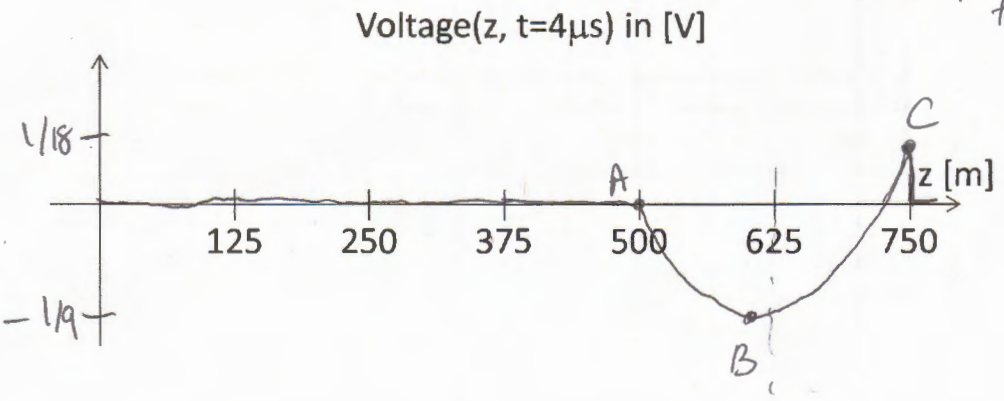
I label 3 points and see where they should go.

Plot the voltage on the transmission line versus position (V vs z) at $t = 2\,\mu s$ and at $t = 4\,\mu s$. Be sure to label the vertical scale in each graph.

2μs ⇒ wave moves 500m

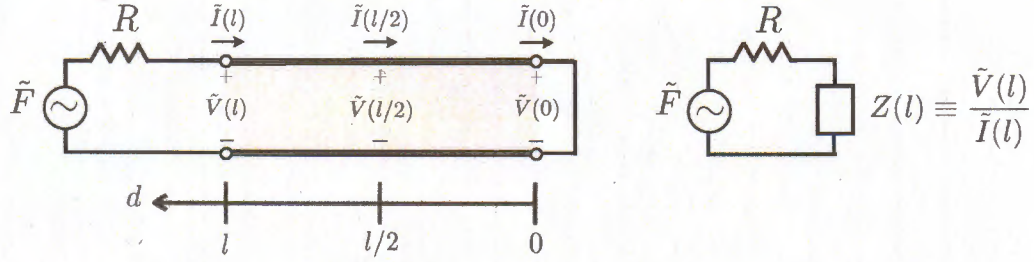


4μs ⇒ wave moves 1000m, i.e. point A makes it to the end and returns $\Gamma_L = -\frac{1}{3}$ so graph flips also



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6. (11 points) Consider a lossless, short circuited, transmission line having length l, characteristic impedance Z_0, and propagation velocity $v = \frac{2}{3}c = 2 \times 10^8$ m/s. As shown in the circuit diagram below, the T.L. is connected in series with a resistor $R = 50\,\Omega$ and an ideal source $f(t) = Re\{\tilde{F}e^{j\omega t}\}$, where $\tilde{F} = 10$ V and $\omega = \pi \times 10^8$ rad/s. The diagram on the right side below depicts the equivalent circuit in terms of the input impedance $Z(l)$ of the shorted line.



a) (2 pts) What is the signal wavelength λ (in m) on the transmission line?

$f = \frac{\omega}{2\pi} = \frac{1}{2}\cdot 10^8$ Hz $\quad \lambda f = v = \frac{2}{3}c \quad \lambda = \frac{2c}{3f} = \frac{2\cdot 10^8}{1/2\cdot 10^8} =$ **4m**

b) If the current through the resistor, denoted $\tilde{I}_R$, is found to equal zero,

i. (2 pt) What is $Z(l)$?

∞

ii. (2 pts) What is the shortest possible length l of the T.L. in terms of the wavelength λ of the given source frequency?

Input open
Load shorted $\Rightarrow$ $l = \frac{\lambda}{4}$ is minimum **$l = 1m$**

iii. (2 pts) Is it possible for $\tilde{V}(l/2) = 0$? If so, what is the shortest possible length of the T.L. (in terms of wavelength λ) in that case?

No. If $\tilde{V}(l/2) = 0$, i.e. a short then $\tilde{V}(l)$ would also be zero, i.e. a short
and so $\tilde{I}_R$ cannot be zero.

iv. (3 pts) For the same length T.L. as in part (b.ii) above, an inductor L is connected to the input end of the T.L. (i.e., at $d = l$). You observe that the current through the resistor $\tilde{I}_R$ is no longer zero. Is the inductor connected in series with the T.L. or in parallel? Briefly explain your answer.

Inductor must be in parallel because we need a shunt path. If it is placed in series, the impedance would add and still be infinite.

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